

A diagram illustrating the components of a linear regression model. At the top left, the text "prior mean" is written. An arrow points from this text to a vector $\vec{m}_0$. To the right of $\vec{m}_0$, an arrow points from the text "prior covariance" to a matrix S_0 . Below $\vec{m}_0$, an arrow labeled f points to a circle containing the symbol $\hat{w}$. Below $\hat{w}$, an arrow labeled f points to a circle containing the symbol y . To the left of y , an arrow labeled f points from the input vector $\vec{x}$ to y . To the right of y , an arrow labeled σ^2 points to a circle containing the symbol ϵ .

$$= -\frac{1}{2\sigma_e^2} (\vec{y} - \underline{X}\vec{\theta})^T (\vec{y} - \underline{X}\vec{\theta}) - \frac{1}{2} (\vec{\theta} - \vec{m}_0)^T \underline{S}_0^{-1} (\vec{\theta} - \vec{m}_0)$$

Aside] If $P(\vec{\theta}) \sim N(\vec{m}_0, \underline{S}_0)$ and $\vec{\theta} \in \mathbb{R}^D + \underline{S}_0 \in \mathbb{R}^{D \times D}$

$$PDF = \frac{1}{\sqrt{2\pi}} \det(\underline{S}_0)^{-\frac{1}{2}} \exp\left(-\frac{1}{2} (\vec{\theta} - \vec{m}_0)^T \underline{S}_0^{-1} (\vec{\theta} - \vec{m}_0)\right)$$

(1) (2) (3)

log posterior

$$\begin{aligned} &= -\frac{1}{2\sigma_e^2} (\vec{y} - \underline{X}\vec{\theta})^T (\vec{y} - \underline{X}\vec{\theta}) - \frac{1}{2} (\vec{\theta} - \vec{m}_0)^T \underline{S}_0^{-1} (\vec{\theta} - \vec{m}_0) \\ &= -\frac{1}{2} \left(\underbrace{\frac{1}{\sigma_e^2} \vec{y}^T \vec{y}}_{\text{stuff}} - \underbrace{\frac{2}{\sigma_e^2} \vec{y}^T \underline{X} \vec{\theta}}_{(2)} + \underbrace{\frac{1}{\sigma_e^2} \vec{\theta}^T \underline{X}^T \underline{X} \vec{\theta}}_{(1)} + \underbrace{\vec{\theta}^T \underline{S}_0^{-1} \vec{\theta}}_{(1)} - 2 \underbrace{\vec{m}_0^T \underline{S}_0^{-1} \vec{\theta}}_{(2)} + \underbrace{\vec{m}_0^T \underline{S}_0^{-1} \vec{m}_0}_{\text{stuff}} \right) \\ &= -\frac{1}{2} \left(\underbrace{\vec{\theta}^T \left(\frac{1}{\sigma_e^2} \underline{X}^T \underline{X} + \underline{S}_0^{-1} \right) \vec{\theta}}_{\text{Quadratic in } \vec{\theta}} - 2 \underbrace{\left(\frac{1}{\sigma_e^2} \vec{y}^T \underline{X} + \vec{m}_0^T \underline{S}_0^{-1} \right) \vec{\theta}}_{\text{linear in } \vec{\theta}} + \underbrace{\text{stuff}}_{\text{constant}} \right) \end{aligned}$$

- Now we have a quadratic in $\vec{\theta}$, which means we may be able to write this as a single Gaussian with some new mean + covariance.

$\Rightarrow$ If we manipulate the above, we can show that the above is equivalent to

$$-\frac{1}{2} (\vec{\theta} - \vec{m}_N)^T \underline{S}_N^{-1} (\vec{\theta} - \vec{m}_N) + \text{const}$$

where $\vec{m}_N = \underline{S}_N \left(\underline{S}_0^{-1} \vec{m}_0 + \frac{1}{\sigma_e^2} \underline{X}^T \vec{y} \right)$

$$+ \underline{S}_N = \left(\underline{S}_0^{-1} + \frac{1}{\sigma_e^2} \underline{X}^T \underline{X} \right)^{-1}$$

$\Rightarrow$ If we have Normal/Gaussian prior + likelihood, the posterior is also Gaussian.

$$\Rightarrow P(\vec{\theta}|\vec{y}) = N(\vec{m}_N, \Sigma_N)$$

Regularization

- When we solve the minimization problem, we typically have

$$\min_{\vec{\theta}} \frac{1}{2} (\vec{y} - f(\vec{\theta}))^T (\vec{y} - f(\vec{\theta}))$$

• Regularization is a way to "penalize" the fitting procedure for certain criteria.

• Ex) Suppose $\vec{\theta} \in \mathbb{R}^D$, $D \gg 1$. If I do NOT want all D parameters to be used, but I just want the most important ones to be fit

• We can introduce regularization

$$\min_{\vec{\theta}} \underbrace{\frac{1}{2} (\vec{y} - f(\vec{\theta}))^T (\vec{y} - f(\vec{\theta}))}_{\text{error in } y \text{ vs. } f(\vec{\theta})} + \underbrace{\frac{1}{2} \sum_{i=1}^D \theta_i^2}_{\text{penalize for } \theta_i > 0}$$



